

Sinchal Kar

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SUMMARY

Web Development Intern at Axlero Innovative Solutions and 2026 B.Tech CSE graduate (KIIT). Programs in Java, JavaScript, Python and SQL; builds full-stack apps with React, Node.js, Express and MongoDB. Shipped a live real-time collaborative editor; strongest in REST API design, database indexing and testing under concurrent load.

EXPERIENCE

Axlero Innovative Solutions

Jul 2026 – Present

Web Development Intern, Technology & Product Development

Remote

Project: SyncSpace – real-time collaborative whiteboard & code editor Live: sync-space-client-gray.vercel.app | [GitHub](#)

- Took SyncSpace from specification to live deployment (React on Vercel, Node.js/Express API on Render); concurrent edits merge conflict-free via Yjs CRDTs over WebSockets, with Socket.io for rooms and chat.
- Designed two-tier MongoDB persistence: debounced binary snapshots for fast room loads plus a schema-enforced append-only update log, so rooms survive server restarts and any session can be replayed.
- Secured rooms with JWT authentication and a 6-level role hierarchy enforced server-side, and built sandboxed code execution for 7 languages including Java (container per run, no network, capped CPU, memory and processes).
- Tested the stack with Vitest and Supertest suites against in-memory MongoDB plus Playwright end-to-end tests driving two live browsers, all run by GitHub Actions CI on every push.

PROJECTS

Skill Sphere: Skill-Based Recruitment Portal

React.js, Node.js, Express, MongoDB, JWT, Nodemailer

- Designed 6 MongoDB collections supporting 500+ applications per posting; cut dashboard query time from 820 ms to 140 ms (83%) with compound indexes on skill tags and application status.
- Load-tested the API at 200 concurrent users, sustaining 180 ms p95 latency with zero failed requests.
- Implemented stateless JWT authentication with bcrypt hashing and role-based access for recruiters and candidates, plus status-change emails with retry-on-failure delivery.

E-Medico: Doctor Appointment System

React.js, Node.js, Express, MongoDB, JWT

- Eliminated double-booking with atomic conditional MongoDB updates for slot locking, verified at 100% booking consistency across 1,000+ simulated parallel requests.
- Sustained 300+ concurrent booking requests at under 250 ms p95 across 50+ doctor profiles and 2 user roles.
- Cut the booking flow from 6 steps to 3 with filtered doctor search by specialisation, availability and location.

Credit Risk & Fraud Detection System

Python, Flask, scikit-learn, Pandas, Docker

- Built a Flask REST API for credit-risk classification with single-record and batch CSV inference, session-based auth and an analytics dashboard; containerised with Docker Compose and a health-check endpoint.
- Trained a Gradient Boosting classifier on a 1,000-record, 20-attribute credit dataset: 91% test accuracy vs. a 70% majority-class baseline, with precision 0.88 / recall 0.85 / F1 0.86 given the 70/30 imbalance.

TECHNICAL SKILLS

Languages: Java, JavaScript, Python, SQL, HTML, CSS

Frontend: React.js, Next.js, Zustand, Vite, Konva, Monaco Editor

Backend: Node.js, Express, Flask, REST API design, Socket.io, WebSockets, Yjs (CRDTs), JWT, RBAC

Databases: MongoDB (Mongoose, schema design, compound indexing, aggregation, atomic updates), SQL

Testing: Vitest, Supertest, Playwright (E2E), Postman, load and concurrency testing

DevOps & Cloud: Git, GitHub Actions (CI), Docker, Docker Compose, Vercel, Render, AWS

Core CS: Data Structures & Algorithms, OOP, DBMS, Data Modelling, Machine Learning (scikit-learn, Pandas)

EDUCATION

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar

2022 – 2026

B.Tech, Computer Science and Engineering | CGPA: 7.59 / 10

Bina Mohit Memorial School, West Bengal

Class XII: 84% (2022) | Class X: 91% (2020)

CERTIFICATIONS

IBM: Web Development Fundamentals | IBM: Generative AI: Introduction and Applications

Infosys Springboard: AWS Certified Developer course